

Vishesh Saluja

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Education

George Mason University , MS in Computer Science <i>Fairfax, VA, USA</i>	May 2025
Guru Gobind Singh Indrapratha University , B.Tech in Information Technology <i>New Delhi, India</i>	GPA: 8.7/10.0 Jul 2023

Experience

Software Developer & Research Assistant , <i>Geoweaver – NASA/ESIP</i>	Feb – Oct 2024
- Led 4 interns on ML workflows for geospatial data and co-authored their first publications. - Added support for PostgreSQL, MySQL, and H2 with CLI integration. - Implemented an in-app code editor and improved UI responsiveness by 30%. - Reduced processing time by 25% in Snow/Ozone Estimation models. - Ran large-scale workflows on Hopper supercomputer with 20% speed-up. - Introduced Vitest for unit testing, significantly accelerating development feedback loops and reducing CI runtime.	
Data Science Intern , <i>QBurst Technologies</i>	Jul – Aug 2022
- Improved BigQuery query performance by 15%. - Conducted training and documentation improving team output by 10%.	
ML Intern , <i>DynPro</i>	Apr – Jun 2022
- Built models improving accuracy by 6.5%. - Delivered regression and clustering results boosting success rate by 20%.	
Software Engineer Intern , <i>Accops Systems</i>	Jan – Mar 2022
- Worked on backend logic using Python and C++. - Reduced bug resolution time by 15%.	
Cloud Trainee , <i>Tech Explica</i>	Jul – Aug 2021
- Deployed multiple websites on AWS. - Increased deployment efficiency by 20%.	

Publication

- Socioeconomic and Industrial Factors Influencing U.S. Ozone Pollution – NASA ESIP (AGU).
- Snow Water Equivalent Estimation Using ML & Remote Sensing – NASA ESIP.

Projects

Mason Critique – <i>Awarded 2nd Place</i>	HackFax 2025
- Developed a full-stack web app using React(Node.js, TypeScript), Firebase, and Tailwind CSS, enabling 50+ GMU students to review courses and professors with real-time feedback. - Engineered backend with Firestore & Google OAuth, integrating GPA calculators, grade charts (Chart.js).	
MindMate – <i>AI Mental Wellness Companion</i>	Feb – Mar 2024
- Built a full-stack mental health app in 36 hours using React, FastAPI, and Azure OpenAI (GPT-4o) to deliver empathetic chat, journaling, and mood-aware prompts while deploying a stable MVP with GPT-driven follow-up logic. - Integrated Framer Motion UI and SQLite backend, enabling 95% response accuracy and storing 100+ emotional logs.	
Diabetic Retinopathy Detection – <i>ML on Retinal Images</i>	Apr – May 2023
Used EfficientNet-B4, ResNet18, and custom CNN with QWK: 0.69.	
Brain Hemorrhage Detection – <i>Binary ResNet Classifier</i>	Dec 2022
Built ResNet18 to classify hemorrhagic vs normal CT scans.	

Skills and Tools

Languages: Python, C, C++, SQL, Java, Bash Libraries/Tools: PyTorch, TensorFlow, Pandas, NumPy, Docker, Kubernetes, GitHub Actions, Terraform	Web & Cloud: HTML, CSS, JS, React, Angular, jQuery, AWS, Azure, GCP, ML/DL, Security
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Extracurricular Activities

- IEEE and Google Developer Student Club member.
- International Taekwondo Medalist.
- Student of the Year (3x).